backbones & *skeletons*

introduction to *network analysis* (*ina*)

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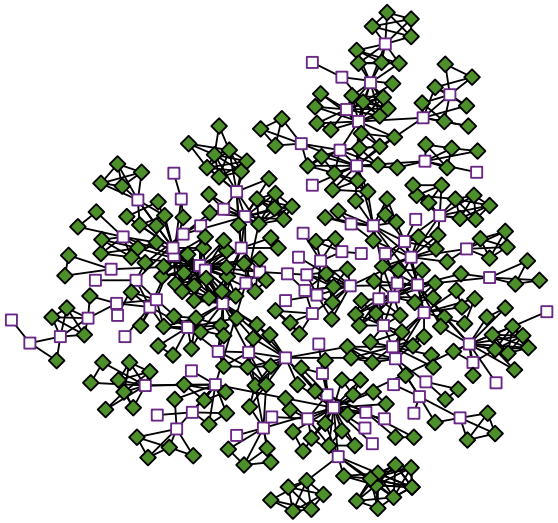
skeletons *overview*

— *network backbone* retains *strongest links/information flow*

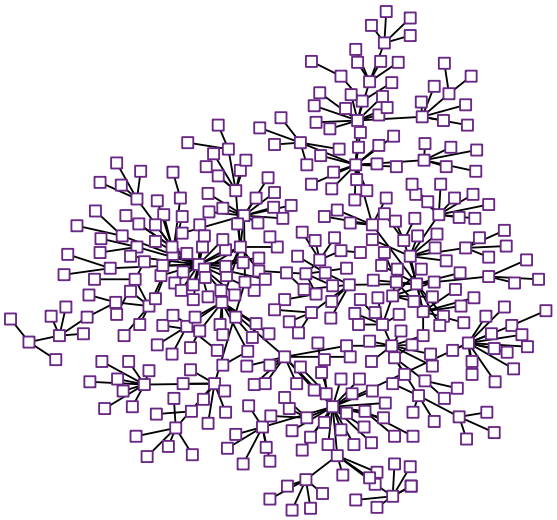
sparsification technique that removes as many links as possible

— *network skeleton* retains *micro/meso/macro structure* [CN17]

simplification technique that retains as many links as possible



coauthorship network

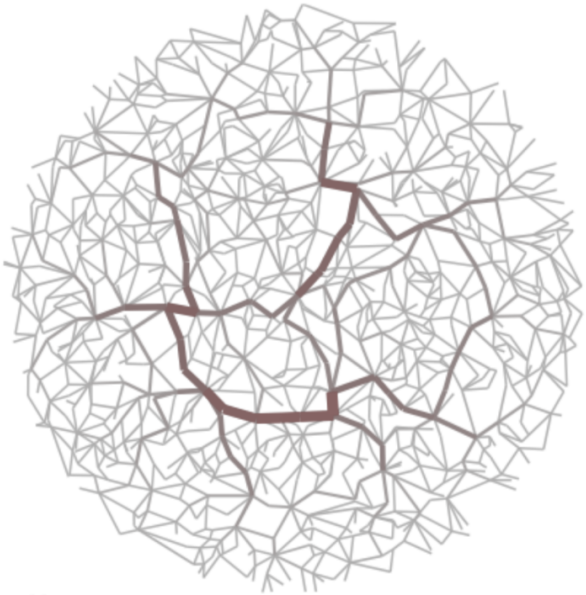


network spanning tree

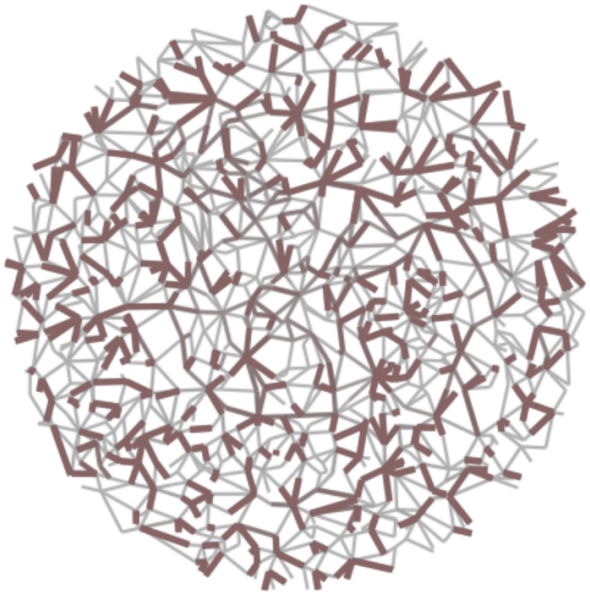
skeletons *geodesics*

— *betweenness backbone* & *salience skeleton* of *graph* [GTB12]

betweenness/salience $\sim$ number of geodesics/spanning trees including edge



high-betweenness backbone

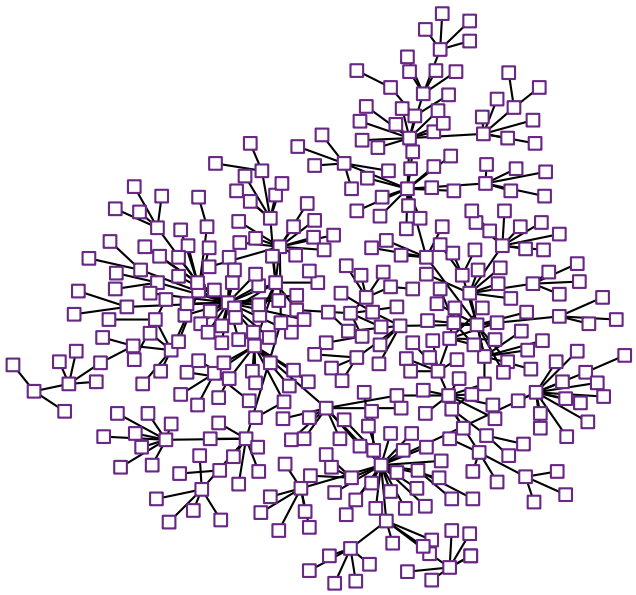


high-salience skeleton

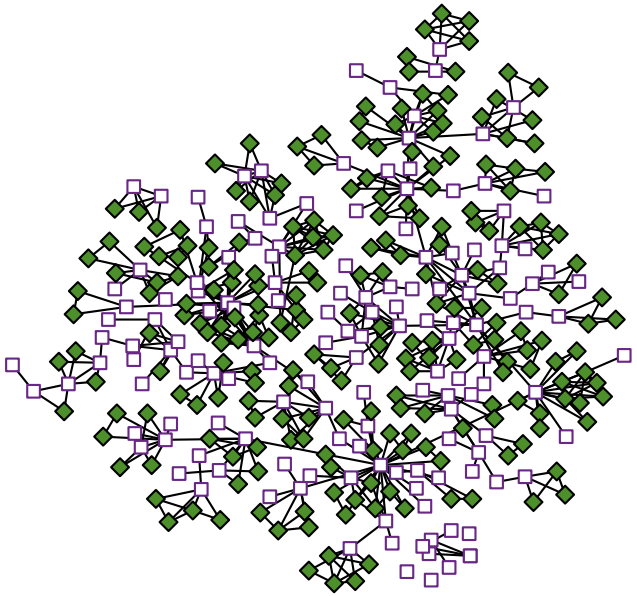
skeletons *convexity*

— *convex skeleton* of famous *network scientists* [MŠ18, Šub18]

largest skeleton where every subgraph includes most geodesics



network spanning tree



convex skeleton $\sim$ tree with cliques

skeletons *statistics*

— *large-scale statistics* of *skeletons of networks* [Šub18]

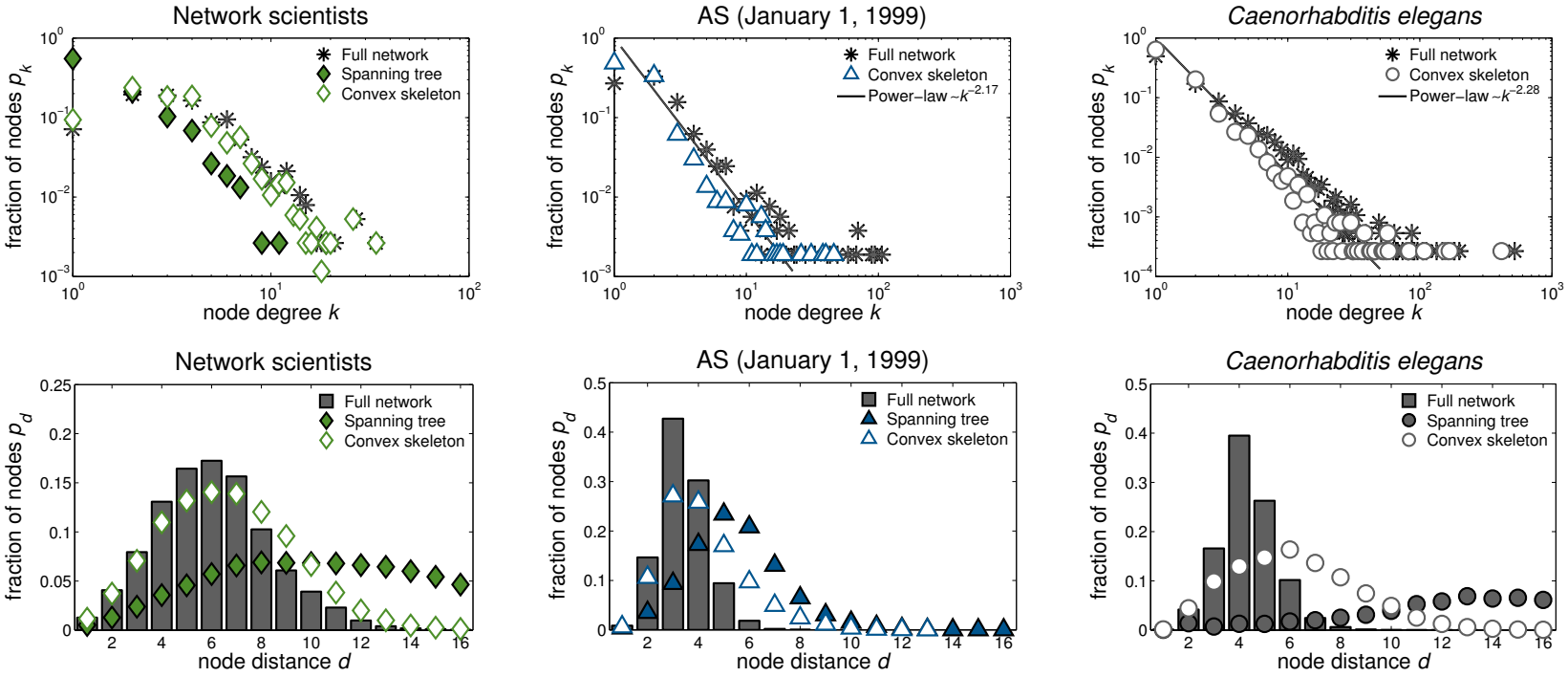
convex skeletons retain connectivity, clustering and geodesics

	clustering $\langle C \rangle$			geodesics $\langle \sigma \rangle$			convexity Xs		
	N	CS	ST	N	CS	ST	N	CS	ST
Jazz musicians	0.62	0.81	0.00	9.71	1.97	1.00	0.12	0.84	1.00
Network scientists	0.74	0.75	0.00	2.66	1.47	1.00	0.85	0.95	1.00
Computer scientists	0.48	0.54	0.00	4.08	1.42	1.00	0.64	0.95	1.00
Plasmodium falciparum	0.02	0.07	0.00	3.71	1.77	1.00	0.43	0.95	1.00
Saccharomyces cerevisiae	0.07	0.10	0.00	2.58	1.19	1.00	0.68	0.88	1.00
Caenorhabditis elegans	0.06	0.12	0.00	6.79	3.03	1.00	0.56	0.85	1.00
AS (January 1, 1998)	0.18	0.21	0.00	3.87	2.32	1.00	0.66	0.91	1.00
AS (January 1, 1999)	0.18	0.27	0.00	3.54	2.05	1.00	0.49	0.95	1.00
AS (January 1, 2000)	0.20	0.25	0.00	4.81	3.07	1.00	0.59	0.90	1.00
Little Rock Lake	0.32	0.69	0.00	22.13	4.32	1.00	0.02	0.82	1.00
Florida Bay (wet)	0.33	0.79	0.00	9.17	1.37	1.00	0.03	0.92	1.00
Florida Bay (dry)	0.33	0.82	0.00	9.37	1.65	1.00	0.03	0.93	1.00

skeletons *distributions*

— *node distributions* of *skeletons of networks* [Šub18]

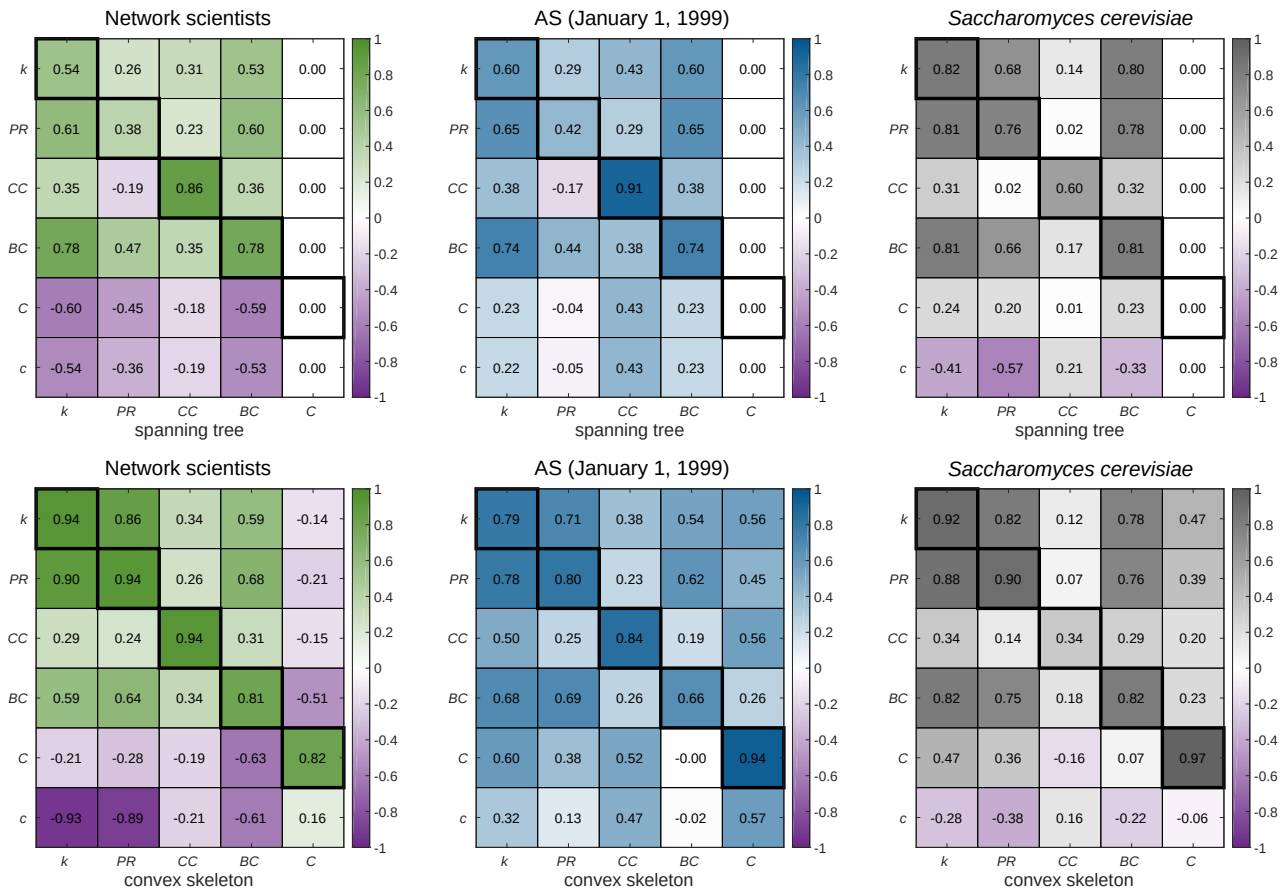
convex skeletons retain node degrees and distances



skeletons *centralities*

— *node centralities* in *skeletons of networks* [Šub18]

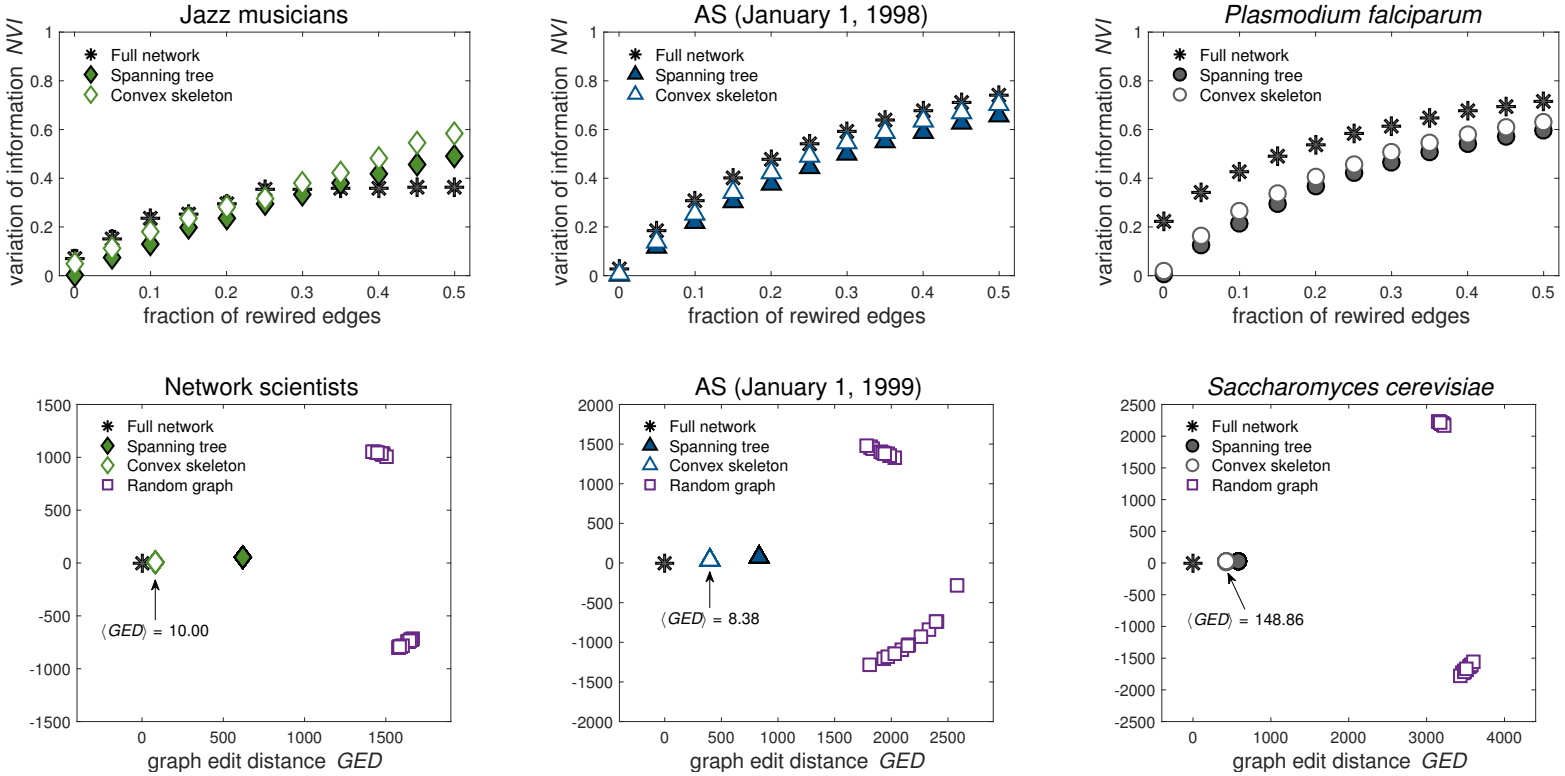
convex skeletons retain node centralities and clustering



skeletons *communities*

— *communities & robustness* of *skeletons of networks* [Šub18]

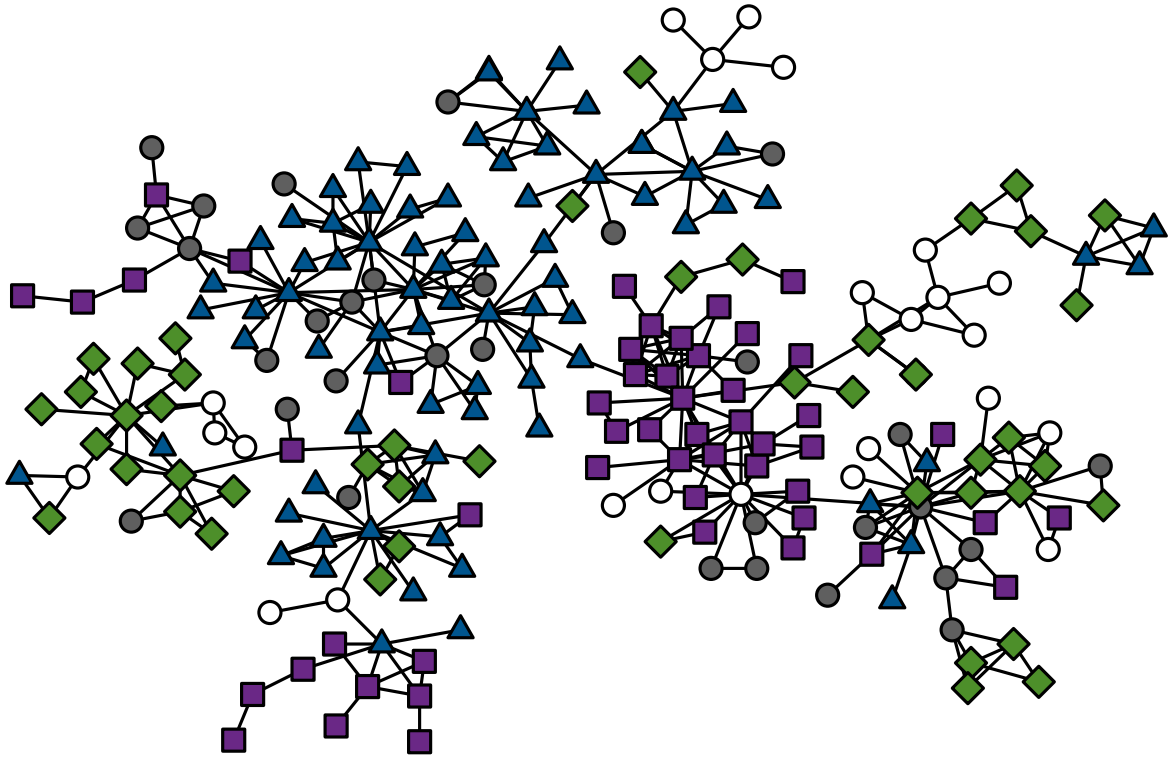
convex skeletons highlight communities with short edit distance



skeletons *coauthorships*

— *convex skeleton* of SI *computer scientists* [Šub18, ŠFCK19]

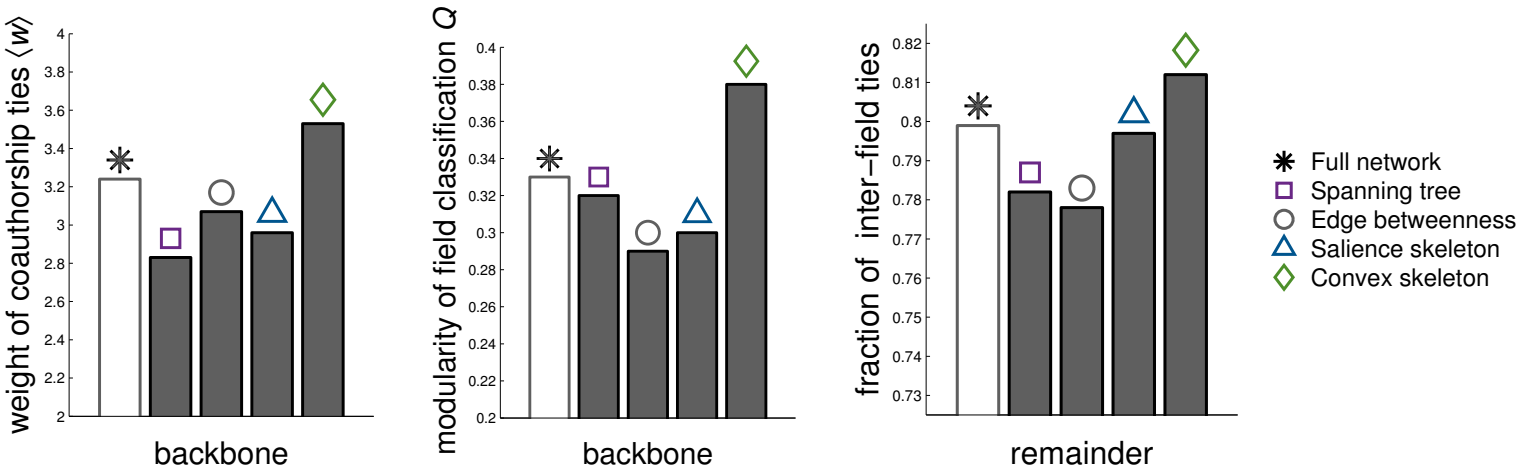
computer theory (◆), information syst. (■), intelligent syst. (▲), programming (○) etc.



skeletons *properties*

— *properties* of *skeletons of coauthorships* [Šub18, ŠFCK19]

convex skeletons retain strong intra-field coauthorships



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